

Assignment 3: Phonon bandstructure

Spring 2026

1. **Phonon bandstructure and heat capacity of 1D diatomic chain.** Consider a hypothetical 1-D diatomic chain with alternating boron and nitrogen atoms, with B-N separation of 1.45 Å (0.145 nm). The atoms are constrained to move only in the direction parallel to the chain, and the spring constant between the atoms is $k = 300$ N/m. In the 1-D nearest-neighbor model discussed in class, the phonon modes are determined by the dynamical equation

$$\begin{pmatrix} 2k - m_B \omega^2 & -k(1 + e^{iqa}) \\ -k(1 + e^{iqa}) & 2k - m_N \omega^2 \end{pmatrix} \begin{pmatrix} u_B \\ u_N \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

where m_B and m_N are the boron and nitrogen atomic masses, q is the phonon wavevector, $\omega(q)$ is the phonon frequency, and u_B and u_N are the boron and nitrogen atom displacements. For the problem below, consider a chain of 201 unit cells with periodic boundary conditions, such that the first Brillouin zone consists of 201 equally spaced points in q (one at the center, and 100 each in the $+q$ and $-q$ directions). For the sections below, use matlab or python to calculate the relevant quantities numerically at each of the q points.

- (a) Solve the dynamical equation numerically at each q value by setting the determinant of the dynamical matrix to zero. Plot the resulting dispersion relation $\epsilon(q) = \hbar\omega(q)$. Use units of electron volts (1 eV = 1.6×10^{-19} J). Check that these points follow the analytical solution derived in class. Also note that you will have two solutions at each point, corresponding to the acoustic and optical branches.

- (b) At a temperature T , the occupancy of a mode with energy ϵ is given by Bose-Einstein statistics

$$N(E, T) = \frac{1}{1 - e^{\epsilon/k_B T}}$$

where k_B is Boltzmann's constant, and the energy in a given mode is given by $E(q, T) = N(\epsilon, T) \epsilon(q)$. Choose 4 values of q (equally spaced) for each branch and plot $N(T)$ and $E(T)$ from 1 – 300 K for each one. A convenient energy conversion is that the thermal energy ($k_B T$) corresponding to room temperature (300 K) is 25 meV. You should see that for $k_B T \gg \epsilon$, $N \approx k_B T / \epsilon$, which is the classical result.

(c) The phonon dispersion can be used to calculate thermodynamic quantities such as the heat capacity. The total energy in the phonon system is determined by summing the energy over all phonon modes

$$U_{ph}(T) = \sum_{branches} \sum_q E(q, T)$$

The specific heat capacity is then given by:

$$C(T) = \frac{1}{M} \frac{dU}{dT}$$

where M is the total mass of the system. Plot C(T) for the diatomic chain, from 1 - 300 K.

2. **Phonon bandstructure of graphene** As discussed in class, the spring constant tensor between the A atom and the B1 atom is given by:

$$K^{A,B1} = \begin{pmatrix} \phi_r & 0 & 0 \\ 0 & \phi_{ti} & 0 \\ 0 & 0 & \phi_{to} \end{pmatrix}$$

- (a) For nearest-neighbors, the in-plane longitudinal force constant $\phi_r = 365$ N/m, the in-plane transverse force constant is $\phi_{ti} = 245$ N/m, and the out-of-plane transverse force constant is $\phi_{to} = 98$ N/m. Following the procedure outlined in the class notes, use these force constants to numerically calculate the phonon frequencies for graphene, for 101 evenly spaced q-points along the $\Gamma - K$ direction, and plot the results.
- (b) Now, you will calculate the specific heat capacity of graphene and graphite. However, the dispersion calculated in (a) is not sufficient to derive the heat capacity alone, since it is a line cut along the $\Gamma - K$ direction not the full 2D Brillouin zone picture. Using the provided phonon density of states data, calculate C(T) for graphene and graphite from 1 - 1000K. Interpolate as needed. Total energy is given by:

$$U(T) = \int_0^\infty g(\omega) \hbar \omega \left(N(E, T) + \frac{1}{2} \right) d\omega$$

where $g(\omega)$ is the phonon DOS, and $N(E, T)$ is the Bose-Einstein distribution. Normalize the given phonon DOS data so that,

$$\int_0^{\infty} g(\omega) d\omega = 6$$

corresponding to 3 branches and 2 atoms per unit cell. Then, the heat capacity will be given as

$$C(T) = \frac{1}{2m_c} \frac{dU}{dT}$$

where m_c is the mass of a carbon atom.